

🧠 PRODUCT REQUIREMENTS DOCUMENT (PRD)

Product Name: QuantTrade Life (QLife)

Category: AI-Powered Gamified Financial Intelligence Platform

1. 🎯 Product Vision

Build a **character-driven financial life simulator** where users:

- Live a financial life (not just play a game)
- Make real-world financial decisions
- Learn through consequences
- Build wealth across life stages

Positioning:

"The Sims + Duolingo + AI Hedge Fund Mentor" for Finance

2. 🧑 Core Concept: Character-Based Financial Lifecycle

Player = Character

Each user creates:

- Avatar (age, background, goals)
- Financial starting condition

Life Stages System (Core Engine)

Stage	Age	Gameplay Focus	Systems
Student	<18	Awareness	Earning basics
College	18–25	First income	Budgeting + investing
Early Career	25–35	Growth	Portfolio building
Family	35–45	Stability	Risk management
Mastery	45+	Optimization	Advanced strategies

👉 Each stage unlocks:

- New financial instruments
- New risks
- New AI behaviors

3. 🎮 Core Gameplay Loop

Earn → Spend → Invest → Learn → Adapt → Grow → Repeat

Detailed:

1. Earn income (job / gig)
2. Allocate funds
3. Invest/trade
4. Market + life events occur
5. Portfolio changes
6. AI feedback
7. Character evolves

4. 💰 **Financial Engine (Simulation Core)**

Asset Classes

- Stocks
- ETFs
- Bonds
- Crypto
- Cash
- Real estate (Phase 2)

Portfolio System

- Allocation %
- Risk score
- Volatility
- Return tracking
- Diversification index

Income Engine

- Salary progression
- Promotions
- Job switching
- Passive income

Expense Engine

- Rent
- Lifestyle choices
- Emergencies

5. 🧠 **AI SYSTEM ARCHITECTURE (Your Core Differentiator)**

A. AI Financial Mentor

- Personalized coaching
- Explains every decision

B. AI Behavior Engine (CRITICAL MOAT)

Tracks:

- Overtrading
- Panic selling
- Risk appetite
- Decision consistency

Outputs:

- Behavioral score
- Psychological profile

C. AI Scenario Generator

Creates:

- Economic cycles
- Personal events
- Market shocks

D. AI Market Engine

- Uses:
 - Polygon.io
 - Finnhub
- Hybrid:
 - Real data + simulated futures

E. AI Community Moderator Bot

- Toxicity detection
- Financial misinformation filtering
- Auto-response assistant

6. 🤖 Story & Scenario Engine (Game Core)

Dynamic Narrative System

Each user has:

- Unique financial journey
- AI-generated story progression

Scenario Examples

1. First Paycheck

- Allocate money
- Learn budgeting

2. Market Crash

- Portfolio drops 30%
- AI teaches risk

3. Startup Opportunity

- High-risk decision

4. Family Responsibility

- Increased expenses

5. Black Swan Event

- Random shock

7. 👤 Community Layer (Retention Engine)

Features

1. Global Feed

- Users share:
 - portfolios
 - strategies
 - insights

2. Stage-Based Communities

- Student
- Early career
- Advanced

👉 This ensures relevance

3. AI Community Bot

Functions:

- Answer questions
- Suggest strategies
- Prevent spam/scam

4. Strategy Marketplace (Phase 2)

- Users publish strategies
- Others follow

8. 🎨 UI/UX SYSTEM (HIGH-END DESIGN)

Design Philosophy

- Dark mode (institutional)
- Glassmorphism

- Neon accents
- Data-first UI

Core Screens

1. Dashboard

- Net worth
- Portfolio
- AI insights

2. Character Screen

- Age progression
- Financial health
- Behavior profile

3. Investment Screen

- Trading charts
- AI suggestions

4. Scenario Screen

- Story-based decisions

5. Community Hub

- Discussions
- AI moderation

9. 🏠 3D / VISUAL ENGINE

Stack

- Three.js / WebGL
- React Three Fiber

Components

- Character avatar (animated)
- Financial world environment
- Event visualization

Visual Feedback Examples

- Portfolio growth → character upgrades
- Loss → environment deterioration

10. 🏗️ TECH ARCHITECTURE

Frontend

- Next.js
- Tailwind
- Zustand / React Query

Backend

- FastAPI
- PostgreSQL (Neon)
- pgvector

AI Stack

- LLM (Claude / Llama / GPT)
- RAG (SEC filings + news)
- Embeddings (MiniLM)

Infra

- Docker
- AWS / Vercel

11. 💰 MONETIZATION

Tier	Offering
Free	Basic gameplay
Pro	AI insights
Elite	Advanced simulations

12. ⚠️ RISKS

Risk	Mitigation
Too complex	Progressive onboarding
Low retention	Daily missions
AI hallucination	RAG system

13. 🔥 KEY PRODUCT DIFFERENTIATION

This is NOT:

- A trading app
- A learning platform

This IS:

A Financial Life Simulator powered by AI

14. 🧠 MASTER BUILD PROMPT (FOR SONNET / GPT / CLAUDE)

Use this to generate system, UI, backend, everything:

🚀 MASTER PROMPT

Build a full-stack AI-powered financial life simulation game called "QuantTrade Life".

Core concept:

- Character-based financial journey (not city-based)
- User progresses through life stages: student → college → career → family → mastery
- Game teaches real-world financial literacy via simulation

Requirements:

1. Gameplay Engine:

- Character system with age progression
- Income, expenses, investments
- Portfolio system (stocks, ETFs, bonds, crypto)
- Scenario engine (life + market events)

2. AI System:

- AI mentor (personalized guidance)
- Behavior tracking (risk, panic, overtrading)
- Scenario generator (dynamic life events)
- Explainable finance insights

3. Community System:

- Stage-based communities
- Global discussion feed
- AI moderation bot
- Strategy sharing system

4. UI/UX:

- Dark theme, glassmorphism, neon highlights
- Dashboard with net worth and portfolio
- Character evolution screen
- Interactive trading interface
- Community hub

5. 3D Engine:

- Use Three.js / React Three Fiber
- Animated character avatars
- Financial world environment
- Visual feedback tied to wealth

6. Backend:

- FastAPI

- PostgreSQL + pgvector
- Real + simulated market data APIs

7. AI Stack:

- LLM integration (RAG with financial data)
- Embeddings for knowledge retrieval

8. Game Loop:

Earn → Spend → Invest → Learn → Adapt → Grow

9. Retention:

- Daily missions
- Achievements
- Leaderboards

10. Output:

- Full system architecture
- Database schema
- API design
- UI wireframes
- Component breakdown
- AI workflow

15. 🧠 Final Strategic Insight

You are sitting on:

👉 A **category-defining product** if executed correctly.

But execution discipline is everything.

Immediate Next Step (Non-Negotiable):

Start with:

- **ONLY Student + College stage MVP**
- No advanced finance initially
- Strong AI mentor from day 1

If you want next:

I can build:

- HLDD (High-Level Design)
- DB schema + APIs
- Exact UI screens (pixel-level)
- AI agent workflows (production-ready)

This is no longer an idea.

This is a **venture-scale product blueprint**.

